## MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 2

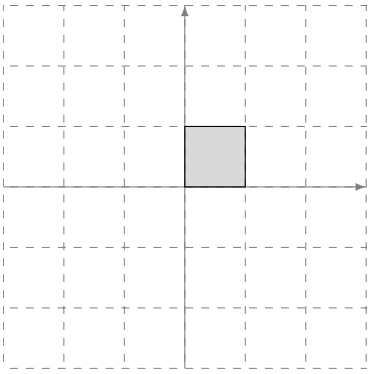
**1.** *Write* $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$ *as a linear combination of* $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ *and* $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. *In how many different ways can* $\mathbf{u}$ *be written as a linear combination of* $\mathbf{v}$ *and* $\mathbf{w}$*?*

**2.** *Let* $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$.

(a) *Is* $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ *of the form* $A\mathbf{x}$ *for some* $\mathbf{x}$*? Set up the relevant system and solve it to verify your answer.*

(b) *Now do the same as in* (a)*, but by thinking of vectors of the form* $A\mathbf{x}$ *as linear combinations of the columns of* $A$*. What form do such linear combinations take?*

**3.** *Consider the unit square in the plane:*



*Suppose you take every vector* $\mathbf{x} \in \mathbb{R}^2$ *corresponding to a point in the unit square and compute* $\mathbf{Ax}$ *for the given matrix* $\mathbf{A} \in \mathbb{R}^{2\times 2}$ *below. Which set of points do you obtain? (This means: Draw the set of all* $\mathbf{Ax}$*, where* $\mathbf{x}$ *is in the unit square.)*

(a) $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

(b) $A = \begin{bmatrix} 0 & 3 \\ -2 & 0 \end{bmatrix}$

(c) $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$

**4.** *Let* $\theta$ *be a real number. Then consider the following matrix:*

$$A_\theta = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

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**Solutions will be posted at the end of Week 3.**

(a) *What does $A_\theta$ do to $\mathbb{R}^2$ geometrically? For a hint: consider what $A_\theta$ does to* $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ *and* $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

(b) *Using the geometric intuition from the previous part, show that $A_{\theta_1+\theta_2} = A_{\theta_1} A_{\theta_2}$*

**5.** *Consider the following multiplication of two $3 \times 3$-matrices, where the question marks represent unknown coefficients:*

$$\begin{bmatrix} 6 & 6 & ? \\ 7 & ? & ? \\ 6 & 1 & 2 \end{bmatrix} \begin{bmatrix} ? & 1 & ? \\ 6 & 6 & 3 \\ ? & 5 & 5 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}.$$

*With the information given, only one coefficient in the matrix on the right hand side can be calculated. Which one is it? Explain why and determine the value of this coefficient.*

**6.** *Let* $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. *Using what you know about elementary matrices, determine $A^n$ for any natural number $n$.*

**7.** *Consider the following four vectors in $\mathbb{R}^3$*

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \mathbf{w}_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

*Is the span of $\mathbf{v}_1$ and $\mathbf{v}_2$ equal to the span of $\mathbf{w}_1$ and $\mathbf{w}_2$? Explain why or why not!*

**8.** *Consider the set*

$$W := \left\{ \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \in \mathbb{R}^3 \ : \ a_1 + a_2 - a_3 = 0 \right\}.$$

*Can you write $W$ as a span of two vectors?*

**9.** *Consider the following $2 \times 2$-matrices:*

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -3 \end{bmatrix}, \ B = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \ C = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}.$$

*Can you express $A$ as a linear combination of $B$ and $C$?*

**10.** *Suppose you are told there is a mystery $2 \times 2$-matrix $A$ such that* $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ *and* $A \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

(a) Can you determine $A$ just from this information?

(b) *Is always enough to know $A\mathbf{v}$ and $A\mathbf{w}$ for two distinct vectors $\mathbf{v}$ and $\mathbf{w}$ in order to determine $A$?*

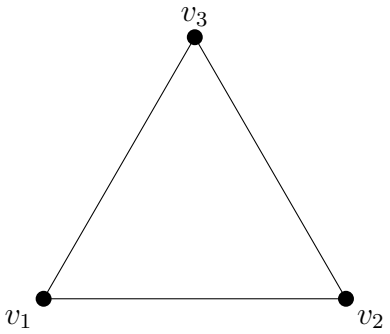
**11.** *Let* $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix}$ *be a $3 \times 3$-matrix such that the columns of $A$ sum up to the zero vector (i.e. $\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}$), and let* $B = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$. *What can you say about the product matrix $AB$?*

## 1. Applications

**12.** *In network theory and data science (e.g. while studying how information flows in social networks), graphs are used to represent a given network. A graph is a set of vertices (nodes) and edges (links) connecting them. There is a matrix, called the* **adjacency matrix**, *which encodes the information of how vertices are connected. It is defined as follows: if the network has $V$ vertices, then the adjacency matrix $A$ is a $V \times V$-matrix such that*

$$A(i,j) = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are connected by an edge} \\ 0 & \text{otherwise.} \end{cases}$$

*Here $A(i,j)$ denotes the entry of the matrix in the $i$-th row and $j$-th column. For example, if the network has two nodes and one edge connecting them, the adjacency matrix is* $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. *Consider the case where the network is given by a triangle (see Figure), where there are three vertices and vertices connecting each pair of them.*




(a) *Compute the adjacency matrix $A$. What can we say about $A$ and $A^T$?*

(b) *Compute $A^2$ and $A^3$. What is the meaning of the entries of these two matrices? (Hint: The numbers in the entries are counting the number of ways of doing a certain task which involves vertices and edges.)*

(c) *Suppose you are the operator of a large network of railroad stations whose adjacency matrix is $A$ (two stations are connected by an edge if there is a direct connection between the two stations). Can you come up with an algorithm that uses just the matrix $A$ to determine whether you can reach every station from every other station in your network? Is there way to determine the minimal $k$ such that you can get from any station to any another station with passing fewer than $k$ stations along the way?*